Different methods for gradient descent stepsize choice

1 Polyak stepsize for gradient descent

1.1 Introduction

The Polyak stepsize is a method for choosing the stepsize for gradient descent (GD). It is an adaptive method for doing so, which means that it allows the algorithm to choose a different stepsize after each step instead of fixing one for the whole process. In the following sections we will look at it in more detail (see Section 1.3).

1.2 GD with a constant stepsize

Let us first recall what GD is. It is an iterative method for finding the minimum of a function. To describe it, let us assume that we are given a function $f\colon \mathbb{R}^d \to \mathbb{R}$ and it is L-smooth, μ-strongly convex, and twice continuously differentiable, i.e., $f \in C^2(\mathbb{R}^d)$*. Then GD can be described as follows.

Algorithm 1 GD

1: **Parameters:** stepsize $\gamma > 0$, starting point $x_0 \in \mathbb{R}^d$
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Set $g_k = \nabla f(x_k)$
4: $x_{k+1} = x_k - \gamma g_k$

Our aim is to choose the stepsize γ so that GD converges "faster". The connection between the stepsize and the convergence speed can be described by the following lemma.

Lemma 1. *Assume f is differentiable, L-smooth and μ-strongly convex. Then the iterates of GD (Algorithm 1) satisfy the following inequality for all $\gamma \geq 0$ and all $k \geq 0$:*

$$\|x_{k+1} - x^*\|^2 \leq (1 - \gamma\mu)\|x_k - x^*\|^2 - 2\gamma(1 - \gamma L) D_f(x_k, x^*), \tag{1}$$

where $x^* \overset{def}{=} \arg\min_{x \in \mathbb{R}^d} f(x)$ and $D_f(x, y)$ is the Bregman divergence, defined as follows:

$$D_f(x, y) \overset{def}{=} f(x) - f(y) - \langle \nabla f(y), x - y \rangle \qquad \forall x, y \in \mathbb{R}^d.$$

Note that $D_f(x, y) \geq 0$ for every convex function f, and hence for every μ-strongly convex function.

*From Algorithm 1 it is convenient that only the differentiability of the function is enough, but for further discussion it is necessary to make stronger assumptions, so we do it here.

Proof of Lemma 1.

$$\begin{aligned}\|x_{k+1} - x^*\|^2 &= \big\|\big(x_k - \gamma\nabla f(x_k)\big) - \big(x^* - \gamma\nabla f(x^*)\big)\big\|^2 \\ &= \big\|(x_k - x^*) - \gamma\big(\nabla f(x_k) - \nabla f(x^*)\big)\big\|^2 \\ &= \|x_k - x^*\|^2 + \gamma^2\|\nabla f(x_k) - \nabla f(x^*)\|^2 - 2\gamma\langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*)\rangle .\end{aligned}$$

From μ-strong convexity we obtain

$$\langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*)\rangle \geq D_f(x_k, x^*) + \frac{\mu}{2}\|x_k - x^*\|^2 .$$

And from L-smoothness we have

$$\|\nabla f(x_k) - \nabla f(x^*)\|^2 \leq 2L\, D_f(x_k, x^*),$$

and as a result,

$$\|x_{k+1} - x^*\|^2 \leq (1 - \gamma\mu)\|x_k - x^*\|^2 - 2\gamma(1 - \gamma L)\, D_f(x_k, x^*),$$

which finalizes the proof. □

As mentioned above, $D_f(x_k, x^*) \geq 0$. Hence, if we take $0 \leq \gamma \leq 1/L$, then $1 - \gamma L \geq 0$ and the last term in (1) is non-positive. Dropping it and applying (1) k times, we obtain the following theorem.

Theorem 2 (Convergence of GD)**.** *Assume that the differentiable function f is L-smooth and μ-strongly convex. Then the iterates of GD with stepsize $0 \leq \gamma \leq 1/L$ satisfy*

$$\|x_k - x^*\|^2 \leq (1 - \gamma\mu)^k \|x_0 - x^*\|^2 . \tag{2}$$

To make this result more concrete, we may ask how many GD iterations it takes for the error to reach a certain target level ε. To answer this, one can start from inequality (2) and show that with $\gamma = 1/L$, for every k that satisfies

$$k \geq \frac{L}{\mu}\log\frac{1}{\varepsilon} \tag{3}$$

we have

$$\|x_k - x^*\|^2 \leq \varepsilon\|x_0 - x^*\|^2 .$$

What we can conclude from Theorem 2 is that if the function is L-smooth and μ-strongly convex, then we can choose some positive $\gamma \leq 1/L$ in **advance** for all iterations of GD, and we will be able to guarantee that after a certain number of steps the squared distance between the current point and the minimizer is reduced by a factor of $\frac{1}{\varepsilon}$ compared to the initial point.

1.3 Adaptive stepsizes: the Polyak rule

Let us now assume that we are allowed to change the stepsize during the algorithm, and pick a new stepsize γ_k at each step k. If we are willing to do this "extra" work, and of course if we are able to do it, we would like to find stepsizes γ_k that are better than a constant one, or at least not worse, in the sense of the ε-guarantee from the previous part (Section 1.2).

It turns out that this is possible, and one such method is called the Polyak stepsize.

Consider a variant of GD (Algorithm 1) with the Polyak stepsize, i.e., with the adaptive (iteration-dependent) stepsize rule

$$\gamma_k = \frac{2D_f(x_k, x^*)}{\|\nabla f(x_k) - \nabla f(x^*)\|^2}, \tag{4}$$

where we assume that $\nabla f(x_k) \neq \nabla f(x^*)$. This is a very natural assumption: since $\nabla f(x^*) = 0$ and the minimizer of a strongly convex function is unique, $\nabla f(x_k) = \nabla f(x^*)$ would mean that $x_k = x^*$, i.e., we have already found the minimum.

(i) From L-smoothness it can be obtained that $\gamma_k \geq 1/L$. That is, the Polyak stepsize is always at least as large as the largest constant stepsize $\gamma = 1/L$ allowed by Theorem 2.

(ii) From μ-strong convexity it can be obtained that $\gamma_k \leq 1/\mu$.

Finally, the Polyak stepsize gives the following one-step guarantee.

Theorem 3 (Polyak)**.** *Assume f is L-smooth and μ-strongly convex, and $\nabla f(x_k) \neq \nabla f(x^*)$. Then one step of GD with the Polyak stepsize γ_k from (4) satisfies*

$$\|x_{k+1} - x^*\|^2 \leq (1 - \gamma_k \mu) \|x_k - x^*\|^2 .$$

Proof of Theorem 3. Let us start from the GD step $x_{k+1} = x_k - \gamma_k \nabla f(x_k)$. Since $\nabla f(x^*) = 0$, we can subtract the zero term $\gamma_k \nabla f(x^*)$ without changing anything, and then group the terms:

$$\|x_{k+1} - x^*\|^2 = \big\|\big(x_k - \gamma_k \nabla f(x_k)\big) - \big(x^* - \gamma_k \nabla f(x^*)\big)\big\|^2 = \big\|(x_k - x^*) - \gamma_k\big(\nabla f(x_k) - \nabla f(x^*)\big)\big\|^2 .$$

Expanding the square and taking γ_k out of the last two terms, we obtain

$$\|x_{k+1} - x^*\|^2 = \|x_k - x^*\|^2 + \gamma_k \Big(\gamma_k \|\nabla f(x_k) - \nabla f(x^*)\|^2 - 2 \langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*)\rangle\Big).$$

Now we bound the inner product from below using μ-strong convexity.

$$\langle x_k - x^*, \nabla f(x_k) - \nabla f(x^*)\rangle \geq D_f(x_k, x^*) + \frac{\mu}{2} \|x_k - x^*\|^2 .$$

The inner product enters with a minus sign and $\gamma_k \geq 0$, so replacing it by this lower bound can only increase the right-hand side. Moreover, in front of the squared gradient norm we substitute the definition (4) of γ_k. This gives

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 + \gamma_k \Bigg(\frac{2D_f(x_k, x^*)}{\|\nabla f(x_k) - \nabla f(x^*)\|^2} \cdot \|\nabla f(x_k) - \nabla f(x^*)\|^2 \\ - 2D_f(x_k, x^*) - \mu \|x_k - x^*\|^2 \Bigg).$$

Here the squared gradient norms cancel, so the first term in the brackets is just $2D_f(x_k, x^*)$, and it cancels with $-2D_f(x_k, x^*)$. Only the term $-\mu \|x_k - x^*\|^2$ remains, and therefore

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - \gamma_k \mu \|x_k - x^*\|^2 = (1 - \gamma_k \mu) \|x_k - x^*\|^2 ,$$

which proves the theorem. □

Applying Theorem 3 at every iteration, we obtain a guarantee for the whole run of Polyak GD.

Corollary 4 (Convergence of Polyak GD)**.** *Let f be L-smooth and μ-strongly convex, and assume that $\nabla f(x_k) \neq \nabla f(x^*)$ for all $k \geq 0$. Then the iterates of GD with the Polyak stepsize (4) satisfy*

$$\|x_k - x^*\|^2 \leq \prod_{i=0}^{k-1} (1 - \gamma_i \mu) \, \|x_0 - x^*\|^2 .$$

Proof of Corollary 4. Theorem 3 bounds the distance after one step by the distance before it. Applying it first to x_k, then to x_{k-1}, and so on down to x_1, we obtain

$$\begin{aligned} \|x_k - x^*\|^2 &\leq (1 - \gamma_{k-1}\mu) \, \|x_{k-1} - x^*\|^2 \leq (1 - \gamma_{k-1}\mu)(1 - \gamma_{k-2}\mu) \, \|x_{k-2} - x^*\|^2 \\ &\leq \cdots \leq \prod_{i=0}^{k-1} (1 - \gamma_i \mu) \, \|x_0 - x^*\|^2 . \end{aligned}$$

□

It remains to compare this product with the rate $(1 - \gamma\mu)^k$ of GD with a constant stepsize.

Corollary 5 (Polyak GD is not slower than GD)**.** *Let $\gamma_0, \ldots, \gamma_{k-1}$ be the Polyak stepsizes from (4), and let $0 < \gamma \leq 1/L$ be any constant stepsize allowed by Theorem 2. Then*

$$\prod_{i=0}^{k-1} (1 - \gamma_i \mu) \leq (1 - \gamma\mu)^k .$$

Proof of Corollary 5. We compare the factors one by one. By (ii), $\gamma_i \leq 1/\mu$, so $1 - \gamma_i\mu \geq 0$. By (i), $\gamma_i \geq 1/L \geq \gamma$, so $1 - \gamma_i\mu \leq 1 - \gamma\mu$. Hence, for every i,

$$0 \leq 1 - \gamma_i \mu \leq 1 - \gamma\mu .$$

Since all the factors are non-negative, multiplying these k inequalities gives the first claim. □

As a consequence, the iteration bound (3) obtained for GD with $\gamma = 1/L$ holds for Polyak GD as well: Polyak stepsizes are at least as good as the best constant stepsize allowed by Theorem 2.